# EXAMINATION QUESTION PAPER

## PG-DAI MARCH 2024

**EXAM Type :** Main Exam/~~Re Exam~~ **Module Name :AI Compute Platforms, Applications & Trends SET:** A

**DATE: 28/01/2025** **Duration: 2Hrs** **Max. Marks: 40**

**Name :** Abhishek Mali

**PRN :** 240840128003

***Instructions:***

***a.*** *Answer these questions sequentially*

***b.*** *Write answer carefully and attach screenshot as much as possible, also number these screenshots as per question number followed by step number.*

***c.*** *For example, Question 1 has multiple screenshots, number them as 1_1, second as 1_2 and so on.*

***d.*** *Kindly ensure before creating zip file, make sure all the required files should be closed to avoid any corruption of the zip file. Uploading only once will be allowed*

***e.*** *Zip file should be renamed with last 4 digit of PRN followed by the exam name. (for example: 240340128001.zip)*

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Q1. Create a repository on Git hub. Add a Readme file to it and initialize. Then get this repository on your local computer. Create a program in any language that takes 2 numbers from user and displays addition of these numbers. Then push this code to the Git hub repository. **[10 Marks]**

Q2. . Create a web site using container (use httpd or nginx image). The website should be accessible on your local computer using URL http://localhost:9100. (Run the website on port 9100). The website should display your name and your PRN number. **[10 Marks]**

Q4. Perform data analysis of the given Amazon sale report dataset using data frames to find - **[20 Marks]**

a. Total revenue generated by each category

b. Sales transactions above 1200

File to be used – Amazon Sale Report.csv.zip

********** ALL THE BEST! **********

# EXAMINATION QUESTION PAPER
## PG-DAI AUGUST 2024

**EXAM Type :** Main Exam/~~Re-Exam~~ **Module Name : Advanced Programming using Python** **SET:** A
**DATE:** 23/09/2024 **Duration: 2Hrs** **Max. Marks: 40**

**Name :** ASHUTOSH AWASTHI

**PRN :** 240840128008

***Instructions:***

*a. All the questions are compulsory.*
*b. The distribution of total marks for a question is given.*
*c. Make sure to properly upload/submit your code file.*
*d. Also, submit the screenshot of output and code for each question.*
*e. Kindly ensure before creating zip file (submit .py files ), make sure all the required files should be closed to avoid any corruption of the zip file.*
*f. Zip file should be renamed with last 4 digit of PRN followed by name of the candidate and exam name (for example: 6004AmitKumar--Python.zip)*
*g. e-Mulyankan Details to upload backup: Link- http://emulyankan.students.acts.in:50*
*UserID- Full PRN*
*Password- Acts123*

**Problem Statement**

Meet Ethan, a data scientist at Renishaw Pune working on a global research project analyzing environmental sensor data. His team gathers massive amounts of data from sensors located all over the world, but before this data can be fed into machine learning models, it needs to be cleaned, processed, and analyzed. Ethan has broken the problem into four tasks, each requiring different skills in string manipulation, regex, lists, dictionaries, numpy, pandas, and exception handling.
Your task is to help Ethan by developing a Python-based solution using Object-Oriented Programming (OOP) concepts.

**Q1 : String Manipulation and Regular Expressions (10 marks)**

Ethan's team receives sensor data in unstructured text format. Your first task is to help clean this data by removing unnecessary characters and standardizing the format.

**Task:**

- Define a class SensorDataProcessor.
- Create a method clean_data(self, data: str) that:
- Removes all special characters (except for spaces).
- Converts the string to lowercase.
- Uses regular expressions to replace all digit sequences with "<num>".

**Sample Input Data**

data = "Sensor_1 recorded: 85 readings on 2023-09-15, at 14:55:01!"

**Sample Output Data**

"sensor recorded num readings on num at num"

**Q2: List and Dictionary Operations (10 marks)**

The cleaned sensor data contains various keywords representing different types of readings. You need to extract meaningful information from these keywords and map them to their corresponding values.

**Task:**
Extend the SensorDataProcessor class by adding a method extract_readings(self, data: str) that:

- Takes a cleaned string as input.
- Uses the keywords ['temperature', 'humidity', 'pressure'] to find values.
- Returns a dictionary with the readings as values. If a reading is missing, return "None" for that reading.

**Sample Input:** data = "temperature: 22 humidity: 55 pressure: num"

**Sample Output:** {'temperature': 22, 'humidity': 55, 'pressure': None}

**Q3: Exception Handling and Numpy Operations (10 marks)**

Ethan's team sometimes encounters corrupted sensor data with invalid entries. You need to ensure that the data is processed correctly, replacing invalid entries with the average of the valid ones.

**Task:**

- Add a method process_numeric_data(self, data: List[float]) that:
- Accepts a list of floating-point numbers.
- Handles exceptions for invalid entries (non-numeric values or None).
- Converts the list to a Numpy array and replaces invalid values with the average of the valid numbers.

**Sample Input :**

data = [22.5, 23.1, None, "invalid", 21.9]

**Sample Output :**

array([22.5, 23.1, 22.5, 22.5, 21.9])

**Q4: Pandas DataFrame Analysis (10 marks)**

Ethan's team tracks multiple sensor readings (e.g., temperature, humidity, pressure) across various time periods. You need to create a method that organizes this data into a Pandas DataFrame and calculates the mean for each type of reading.

**Task:**

- Add a method analyze_data(self, readings: Dict[str, List[float]]) that:
- Converts the input dictionary into a Pandas DataFrame.
- Appends a row with the mean of each column at the bottom.

**Sample Input :**

```
readings = {
    'temperature': [22.1, 23.0, 21.9],
    'humidity': [55, 60, 57],
    'pressure': [1012, 1010, 1011]
}
```

**Sample Output :**

```
   temperature  humidity  pressure
0        22.1        55      1012
1        23.0        60      1010
2        21.9        57      1011
3        22.33     57.33     1011  # Mean values
```

**Notes:**

- Your solution should use **OOP principles** for better modularity and encapsulation.
- Handle exceptions for all cases, especially during invalid data entry and processing.
- Use **numpy** for numerical operations and **pandas** for data analysis.

Good luck! Ethan is counting on you to help his team efficiently process and analyze the sensor data at Renishaw Pune.

********** ALL THE BEST **********

# EXAMINATION QUESTION PAPER

## PG-DAI AUGUST 2024

EXAM Type : Main Exam/~~Re Exam~~ Module Name : Java Programming SET: A

DATE: 08/10/2024 Duration: 2Hrs Max. Marks: 40

Name : ANURAG P. EKKA

PRN : 240840128007

*Instructions:*

*a. All the questions are compulsory.*
*b. The distribution of total marks for a question is given.*
*c. Make sure to properly upload/submit your code file.*
*d. Also, submit the screenshot of output and code for each question.*
*e. Kindly ensure before creating zip file , make sure all the required files should be closed to avoid any corruption of the zip file.*
*f. Zip file should be renamed with last 4 digit of PRN followed by name of the candidate and exam name (for example: 6004AmitKumar--Java.zip)*
*g. e-Mulyankan Details to upload backup: Link- http://emulyankan.students.acts.in:50*
*UserID- Full PRN*
*Password- Acts123*

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**Q1) Library Management System:** Create a Java program to manage book records in a library, allowing users to perform the following operations:

1. Add a new book with **title, author, publication year, ISBN, borrowed**
2. Display all books in the library.
3. Search for a book by **title** or **author**.
4. Borrow a book by specifying the book's ISBN.
5. Return a borrowed book by specifying the book's ISBN.

**Requirements:**

- **borrowed** is a boolean field, when a book is borrowed its value will be true else false.
- Use an array or collection to store book records.
- Use proper exception handling and validation checks.
- Write clean, readable, and well-commented code.

**Marks:**

| | |
|---|---|
| - Implementation of ***add new book*** | **[10 Marks]** |
| - Implementation of ***display all books*** | **[4 Marks]** |
| - Implementation of ***search by title*** | **[4 Marks]** |
| - Implementation of ***search by author*** | **[4 Marks]** |
| - Implementation of ***Borrow book*** | **[4 Marks]** |
| - Implementation of ***Return borrowed book*** | **[4 Marks]** |
| - Proper exception handling and validation checks | **[4 Marks]** |
| - Output and user interaction | **[4 Marks]** |
| - Code quality, readability, and commenting | **[2 Marks]** |

********** ALL THE BEST **********

# EXAMINATION QUESTION PAPER
## PG-DAI AUGUST 2024

EXAM Type : Main Exam/~~Re-Exam~~ Module Name : Java Programming SET: B
DATE: 08/10/2024 Duration: 2Hrs Max. Marks: 40

Name : S. Abhinav Ram

PRN : 240840128032

*Instructions:*

*a. All the questions are compulsory.*
*b. The distribution of total marks for a question is given.*
*c. Make sure to properly upload/submit your code file.*
*d. Also, submit the screenshot of output and code for each question.*
*e. Kindly ensure before creating zip file , make sure all the required files should be closed to avoid any corruption of the zip file.*
*f. Zip file should be renamed with last 4 digit of PRN followed by name of the candidate and exam name (for example: 6004AmitKumar--Java.zip)*
*g. e-Mulyankan Details to upload backup: Link- http://emulyankan.students.acts.in:50*
*UserID- Full PRN*
*Password- Acts123*

**Q1) Phonebook Program:** Create a Java program to manage a simple phonebook that allows users to perform the following operations:

1. Add a new contact with a name and phone number.
2. Display all contacts in the phonebook.
3. Search for a contact by name and display their phone number.
4. Delete a contact by name.

**Constraints/Requirements:**
- Name is unique and there is only one phone number saved under a name.
- Use an array/collection of objects to store contact information.
- Implement methods for each operation (add contact, display contacts, search contact, delete contact).
- Use proper exception handling and validation checks.
- Write clean, readable, and well-commented code.

**Marks:**

| | |
|---|---|
| - Implementation of ***add contact*** | **[10 Marks]** |
| - Implementation of ***display contact*** | **[7 Marks]** |
| - Implementation of ***search contact*** | **[7 Marks]** |
| - Implementation of ***delete contact*** | **[7 Marks]** |
| - Proper exception handling and validation checks | **[4 Marks]** |
| - Output and user interaction | **[3 Marks]** |
| - Code quality, readability, and commenting | **[2 Marks]** |

********** ALL THE BEST **********

# EXAMINATION QUESTION PAPER
## PG-DHPCSA AUGUST 2024

**EXAM Type :** Main Exam/~~Re-Exam~~ **Module Name : Data Analytics** **SET:** A
**DATE: 28/10/2024** **Duration: 2Hrs** **Max. Marks: 40**

**Name :** Geetika Kanwar

**PRN :** 240840128016

*Instructions:*

a. *All the questions are compulsory.*
b. *The distribution of total marks for a question is given.*
c. *Make sure to properly upload/submit your code file.*
d. *For answers create a word document name it as per your PRN. Answer these questions sequentially. Write answer carefully and attach screenshot as much as possible, also number these screenshots as per question number followed by step number. For example, Question 1 has multiple screenshots, number them as 1_1, second as 1_2 and so on*
e. *Zip file should be renamed with last 4 digit of PRN followed by name of the candidate and exam name (for example: 6004AmitKumar--Python.zip)*
f. *e-Mulyankan Details to upload backup: Link- http://emulyankan.students.acts.in:50*

*UserID- Full PRN*
*Password- Acts123*

***Important Note: Reload dataset for each question***

***Use dataset:*** ***Dog_Bites_Data***

1. Import this dataset into python using appropriate packages. Print top 15 rows for columns of DateOfBite, Breed, Age, SpayNeuter, ZipCode with no missing data and SpayNeuter is "FALSE"?

**[7 Marks]**

2. Remove missing data and draw a colourful plot to depict Correlation between Breed, Gender, SpayNeuter?

**[6 Marks]**

3. Remove missing Data in Age column and perform data cleaning, perform an Outlier detection on Age and create a box-plot for the same. **[7 Marks]**

***Use dataset:*** ***Titanic-Dataset***

1. Find the column which has missing data in it. If yes then for any one column update the missing data with the mean of its respective "Pclass" value. **[5 Marks]**

2. Conduct a Z-test to compare the average fare paid by survivors (Survived = 1) and non-survivors (Survived = 0) in the dataset. Provide an interpretation of the results. **[10 Marks]**

3. Calculate Mean, Median, Mode, Quartile Ranges, Standard Deviation of Age for "Female" passengers?

**[5 Marks]**

********** ALL THE BEST **********

# EXAMINATION QUESTION PAPER
## PG-DHPCSA AUGUST 2024

EXAM Type : Main Exam/~~Re Exam~~ | Module Name : Data Analytics | SET: B
DATE: 28/10/2024 | Duration: 2Hrs | Max. Marks: 40

Name : SONAL SOMYA

PRN : 240840128034

*Instructions:*

a. *All the questions are compulsory.*
b. *The distribution of total marks for a question is given.*
c. *Make sure to properly upload/submit your code file.*
d. *For answers create a word document name it as per your PRN. Answer these questions sequentially. Write answer carefully and attach screenshot as much as possible, also number these screenshots as per question number followed by step number. For example, Question 1 has multiple screenshots, number them as 1_1, second as 1_2 and so on*
e. *Zip file should be renamed with last 4 digit of PRN followed by name of the candidate and exam name (for example: 6004AmitKumar--Python.zip)*
f. *e-Mulyankan Details to upload backup: Link- http://emulyankan.students.acts.in:50*

*UserID- Full PRN*
*Password- Acts123*

***Important Note: Reload dataset for each question***

***Use dataset:*** ***Dog_Bites_Data***

1. 1. Import this dataset into python using appropriate packages. Print top 15 rows for columns of DateOfBite, Age, Gender, SpayNeuter, ZipCode with no missing data and Gender is not "U"? **[7 Marks]**
2. 2. Remove missing data and draw a colourful plot to depict Correlation between Breed, Gender, SpayNeuter? **[6 Marks]**
3. 3. Remove missing Data in Age column and perform data cleaning, perform an Outlier detection on Age and create a box-plot for the same. **[7 Marks]**

***Use dataset:*** ***Titanic***

1. Find the column which has missing data in it. If yes then for any one column update the missing data with the mode of its respective "Pclass" value. **[5 Marks]**
2. Perform a Chi-square test among passenger class (Dataset Column Pclass) and survival (Dataset column Survived). Comment on the inference of the result. **[10 Marks]**
3. Calculate Mean, Median, Mode, Quartile Ranges, Standard Deviation of Age for "Male" passengers? **[5 Marks]**

********** ALL THE BEST **********

**EXAMINATION QUESTION PAPER**
**PG-DAI AUGUST 2024**

**EXAM Type :** Main Exam/~~Re-Exam~~ **Module Name : Deep Neural Networks** **SET:** B
**DATE: 05/12/2024** **Duration: 2Hrs** **Max. Marks: 40**

**Name :** Mayuresh Jaywant Patil

**PRN :** 240840128027

*Instructions:*

a. ***Total Number of Questions:*** 4
b. ***Evaluation Criteria:***
   - ***Marks are allocated as indicated next to each question.***
   - ***Write your analysis and comments for each cell/output.***
   - ***Marks breakdown will depend on the accuracy, completeness, clarity of the solution and your observations for each output.***
c. ***Submission Guidelines:***
   1. ***Save your answers in a*** *single .zip file.*
   2. *File naming format: YourName_RollNumber.zip*
   3. ***Include any code, outputs, notebooks or other files as instructed per question***
d. ***e-Mulyankan Details to upload backup: Link- http://emulyankan.students.acts.in:50***
   ***UserID- Full PRN***
   ***Password- Acts123***

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***Question 1: Neural Network for Regression Task*** [10 Marks]

Build and train a deep neural network using the **California Housing Dataset** to predict the median house value.

1. Preprocess the dataset by normalizing all numerical features.
2. Define a neural network architecture with at least two hidden layers.
3. Determine total number of training parameters of the model.
4. Train the model for 20 epochs and evaluate its performance using the Mean Absolute Error (MAE) metric.

**Expected Output:**

- Code for preprocessing and training the model.
- A summary of the model architecture.
- A table showing the training and validation MAE for all epochs.

***Question 2: Regularization and Performance Improvement*** [15 Marks]

Enhance the model from Question 1 by adding regularization techniques:

1. Apply both **Dropout** and **L2 Regularization**.
2. Experiment with different dropout rates (e.g., 0.1, 0.2, 0.3) and observe their effects on performance.

Train the updated model for 20 epochs and compare the results with the original model.

**Expected Output:**

**EXAMINATION QUESTION PAPER**
**PG-DAI AUGUST 2024**

**EXAM Type :** Main Exam/~~Re-Exam~~ **Module Name : Deep Neural Networks** **SET:** A
**DATE: 05/12/2024** **Duration: 2Hrs** **Max. Marks: 40**

**Name :** Makrand Phanekar

**PRN :** 240840128013

*Instructions:*

a. *Total Number of Questions: 9*
b. *Evaluation Criteria:*
   - *Marks are allocated as indicated next to each question.*
   - *Write your analysis and comments for each cell/output.*
   - *Marks breakdown will depend on the accuracy, completeness, clarity of the solution and your observations for each output.*
c. *Submission Guidelines:*
   1. *Save your answers in a single .zip file.*
   2. *File naming format: YourName_RollNumber.zip*
   3. *Include any code, outputs, notebooks or other files as instructed per question*
d. *e-Mulyankan Details to upload backup: Link- http://emulyankan.students.acts.in:50*
   *UserID- Full PRN*
   *Password- Acts123*

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**Question 1: Neural Network Architecture Design** **[10 Marks]**

Design a feedforward neural network using a deep learning framework of your choice (TensorFlow or PyTorch) to classify images from the **MNIST** dataset.

1. Define a neural network architecture with at least three hidden layers.
2. Choose appropriate activation functions for each layer and justify your choices.
3. Train the model for 10 epochs and report the training and validation accuracies.

**Expected Output:**

- Code for defining and training the network.
- A summary of the model architecture.
- A table showing training and validation accuracies for all epochs.

**Question 2: Regularization Techniques** [15 Marks]

Enhance the neural network from Question 1 by applying the following regularization techniques:

1. **Dropout:** Add dropout layers to reduce overfitting.
2. **L2 Regularization:** Apply weight decay during training.

Train the updated model for 10 epochs and compare the results with the previous model.

**Expected Output:**

- Code implementing dropout and L2 regularization.
- A table comparing training and validation accuracies before and after regularization.
- A brief explanation of how regularization affected the model's performance

[15 Marks]

**Question 3: Model Optimization and Evaluation**

Using the model from Question 2:

1. Apply an optimizer of your choice (e.g., Adam, SGD) and justify your selection.
2. Evaluate the model on the test set using metrics like accuracy, precision, recall, and F1-score.
3. Plot a confusion matrix and interpret the results.

**Expected Output:**

- Code showing the optimizer implementation and evaluation.
- A table showing the evaluation metrics (accuracy, precision, recall, F1-score).
- Confusion matrix plot with interpretation

********** ALL THE BEST **********

**EXAMINATION QUESTION PAPER**
**PG-DAI AUGUST 2024**

**EXAM :** Main Exam/~~Re-Exam~~ **Module Name : Natural Language Processing & Computer Vision SET:** A

**DATE: 01/01/2025** **Duration: 2Hrs** **Max. Marks: 40**

**Name :** Dhananjay Bharat Gaikwad

**PRN :** 240840128012

*Instructions:*

a. *Total Number of Questions : 4*
b. *Create a folder named with your Full PRN Number and save all the work related to this lab exam in the folder*
c. *At the end of the exam, ZIP the folder with the following files inside:*
   i) *Code files (.ipynb or .py)*
   ii) *Any output files (.csv, .png, etc.)*
d. *Ensure your code is well-commented and explains each step clearly.*
e. *Upload the zipped folder using the provided link.*
f. *e-Mulyankan Details to upload backup: Link- http://emulyankan.students.acts.in:50*
   *UserID- Full PRN*
   *Password- Acts123*

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**[10 Marks]**

**Q1.** Image Transformation

1. Read a sample image.
2. Perform the following transformations:
   - Resize the image to **256x256 pixels**.
   - Rotate the image by **45 degrees**.
3. Display the **original image** and the **transformed image** side by side.
4. Save the transformed image to a file.

**[10 Marks]**

**Q2.** Histogram Equalization

1. Read a sample image and convert it to grayscale.
2. Perform **histogram equalization** on the image to enhance contrast.
3. Display the **original grayscale image** and the **histogram-equalized image** side by side.
4. Save the equalized image to a file.

**[10 Marks]**

**Q3.** Text Vectorization

1. Write Python code to preprocess a sample.txt file for NLP tasks:
   - Tokenization
   - Removing punctuation
   - Converting text to lowercase
   - Removing stop words
2. Perform **TF-IDF Vectorization** on the preprocessed text.
3. Save the resulting TF-IDF matrix to a .csv file.

**[10 Marks]**

**Q4.** Sentiment Analysis

1. Load a dataset (Amazon_reviews.csv or equivalent).

2. Preprocess the dataset:
   - Work on "Review Content" column
   - Clean the text (removing special characters, converting to lowercase).
   - Perform tokenization and padding.
3. Build a **binary classification model** for sentiment analysis using one of the following algorithms:
   - Logistic Regression
   - LSTM
   - BERT (optional).
4. Save the predictions to a file (predictions.csv).
5. Evaluate your model and display the **accuracy** and **confusion matrix**.

**Note:** Provide comments and documentation in your code to explain each step and make it easily understandable.

********** ALL THE BEST **********

**EXAMINATION QUESTION PAPER**
**PG-DAI AUGUST 2024**

**EXAM :** Main Exam/~~Re-Exam~~ **Module Name : Natural Language Processing & Computer Vision SET:** B

**DATE: 01/01/2025** **Duration: 2Hrs** **Max. Marks: 40**

**Name :** Dhruva Prabhakar Patil

**PRN :** 240840128026.

*Instructions:*

a. *Total Number of Questions : 4*
b. *Create a folder named with your Full PRN Number and save all the work related to this lab exam in the folder*
c. *At the end of the exam, ZIP the folder with the following files inside:*
   i) *Code files (.ipynb or .py)*
   ii) *Any output files (.csv, .png, etc.)*
d. *Ensure your code is well-commented and explains each step clearly.*
e. *Upload the zipped folder using the provided link.*
f. *e-Mulyankan Details to upload backup: Link- http://emulyankan.students.acts.in:50*
   *UserID- Full PRN*
   *Password- Acts123*

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Q1. Image Filtering **[10 Marks]**

1. Read a sample image.
2. Apply the following filters:
   - **Median filter** with a kernel size of 5x5.
   - **Gaussian filter** with a kernel size of 5x5 and standard deviation of 1.
3. Display the **original image** and the **filtered images** (median and Gaussian) side by side.
4. Save both filtered images to separate files.

Q2. Edge **Detection** **[10 Marks]**

1. Read a sample image and convert it to grayscale.
2. Apply the **Sobel operator** to detect vertical and horizontal edges in the image.
3. Combine the vertical and horizontal edges into a single edge-detected image.
4. Display the **original grayscale image** and the **edge-detected image** side by side.
5. Save the edge-detected image to a file.

Q3. Text Preprocessing and Word Embeddings **[10 Marks]**

1. Write Python code to preprocess a sample.txt file for NLP tasks:
   - Tokenization
   - Removing punctuation
   - Converting text to lowercase
   - Removing stop words
   - Lemmatization
2. Perform **Word2Vec embedding** on the preprocessed text and display the word embeddings for any five unique words.

3. Save the word embeddings to a .csv file. **[10 Marks]**

Q4. Sentiment Analysis

1. Load a dataset IMDB Dataset.csv with two columns:
   - **Review**: Text content of the review.
   - **Sentiment**: Label (Positive/Negative).
2. Preprocess the dataset:
   - Clean the text (removing special characters, converting to lowercase).
   - Perform tokenization and padding.
3. Build a **binary classification model** for sentiment analysis using one of the following algorithms:
   - Logistic Regression
   - Naive Bayes
4. Save the predictions to a file (predictions.csv).
5. Evaluate your model and display the **accuracy** and **confusion matrix.**

**Note:** Provide comments and documentation in your code to explain each step and make it easily understandable.

********** ALL THE BEST **********

**EXAMINATION QUESTION PAPER**
**PG-DAI AUGUST 2024**

**EXAM :** Main Exam/~~Re-Exam~~ **Module Name : AI Compute Platforms, Applications & Trends** **SET:** B

**DATE: 08/01/2025** **Duration: 2Hrs** **Max. Marks: 40**

**Name :** VAN DEIV NAYAK

**PRN :** 240840128036

*Instructions:*

a. *Total Number of Questions : 2*
b. *Create a folder and save all the work related to this lab exam in the folder*
c. *At the end of the exam, ZIP the folder with your last 4 digits of your PRN followed by your name and Exam Name*
d. *Ensure your code is well-commented and explains each step clearly.*
e. *Upload the zipped folder using the provided link.*
f. *e-Mulyankan Details to upload backup: Link- http://emulyankan.students.acts.in:50*
*UserID- Full PRN*
*Password- Acts123*

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**. Q 1.** Do the following tasks : **[20 Marks]**

Use tips dataset.

Write an Apache Spark program to find all tips that are 5% or above the bill amount using Spark DataFrame.

Do the same thing as above, but this time use Spark SQL.

Note - Use tips.csv

Create a Kafka Producer, which will allow us to add a new 'tips' record.

A Kafka Consumer should read this record and display the number of records in the tips dataframe.

Run the Producer three times to add three records and display the updated record count.

Note - use tips_json file for supplying data to Kafka Producer.

**Q.2** Do the following: **[20 Marks]**

Create a Dockerfile to run a Welcome to CDAC printing program using Python and Flask.

Run the above on the local machine using Docker.

Create a GitHub repository and move all the files for this project on to GitHub.

********** ALL THE BEST **********

**EXAMINATION QUESTION PAPER**
**PG-DAI AUGUST 2024**

**EXAM :** Main Exam/~~Re-Exam~~ **Module Name : AI Compute Platforms, Applications & Trends** **SET:** A

**DATE: 08/01/2025** **Duration: 2Hrs** **Max. Marks: 40**

**Name :** Adarsh Sharma

**PRN :** 240840128004

***Instructions:***

a. ***Total Number of Questions : 2***
b. ***Create a folder and save all the work related to this lab exam in the folder***
c. ***At the end of the exam, ZIP the folder with your last 4 digits of your PRN followed by your name and Exam Name***
d. ***Ensure your code is well-commented and explains each step clearly.***
e. ***Upload the zipped folder using the provided link.***
f. ***e-Mulyankan Details to upload backup: Link- http://emulyankan.students.acts.in:50***
***UserID- Full PRN***
***Password- Acts123***

---

. Q 1. Do the following tasks : **[20 Marks]**

Use titanic dataset.

Write an Apache Spark program to find all female passengers who survived using Spark DataFrame.

Do the same thing as above, but this time use Spark SQL.

Note - Use Titanic-Dataset.csv

Create a Kafka Producer, which will allow us to add a new 'titanic' record.

A Kafka Consumer should read this record and display the number of records in the titanic dataframe.

Run the Producer three times to add three records and display the updated record count.

Note - use Titanic-Dataset_json file for supplying data to Kafka Producer.

Q.2 Do the following: **[20 Marks]**

Create a Dockerfile to run a Hello World program using Python and Flask.

Run the above on the local machine using Docker.

Create a GitHub repository and move all the files for this project on to GitHub.

********** ALL THE BEST **********

# EXAMINATION QUESTION PAPER

## PG-DAI August 2024

**EXAM Type :** ~~Main Exam~~/Re-Exam **Module Name :** Deep Neural Networks **SET:** C

**DATE:** 09/01/2025 **Duration:** 2Hrs **Max. Marks: 40**

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**PRN :** 240840128001

***Instructions:***

***a.*** *Create a folder with your full PRN number (for example: 240840128001.zip) and save all the LAB ReEXAM work inside of that folder.*

***b.*** *At the end of the exam, close all the applications and then ZIP the folder to be uploaded (.ipynb /.py / plots / predictions.csv)*

**Q1. Create a Simple ANN (Classification) model for the Titanic Dataset using Tensorflow or pyTorch.** **[40 Marks]**

**Tasks:**

**Data Preparation** [6 Marks]

- Load the titanic dataset: Load the train and test data.
- Preprocess the data: Normalize the input features, and split the dataset into training and testing sets.

**Model Architecture** [8 Marks]

- Design a feedforward neural network architecture.
- Define the number of layers, neurons in each layer, and activation functions.

**Model Training** [10 Marks]

- Initialize the model parameters (weights and biases).
- Define a loss function appropriate for the task.
- Train the model using gradient descent or any suitable optimization algorithm.
- Monitor the training process by observing the loss and accuracy metrics.

**Model Evaluation** [8 Marks]

- Evaluate the trained model on the testing dataset.
- Calculate and report the accuracy.

**Experimentation** [8 Marks]

- Experiment with different hyperparameters such as learning rate, number of hidden layers, and number of neurons in each layer.
- Explore different activation functions (e.g., ReLU, sigmoid, tanh) and observe their impact on the model's performance.

**Deliverables:**

- Python code (ipynb) implementing the neural network.
- Training loss curve plot.
- Evaluation results saved in predictions.csv file.

**Note:** Provide comments and documentation in your code to explain each step and make it easily understandable

********** ALL THE BEST! **********